

TM Forum Component

Bill Generation Management

TMFC030

Maturity Level: General Availability (GA)	Team Approved Date: 10-Dec-2024
Release Status: Production	Approval Status: TM Forum Approved
Version 2.2.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2025. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the [TM FORUM IPR Policy](#), must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an “AS IS” basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

TM FORUM invites any TM FORUM Member or any other party that believes it has patent claims that would necessarily be infringed by implementations of this TM Forum Standards Final Deliverable, to notify the TM FORUM Team Administrator and provide an indication of its willingness to grant patent licenses to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this deliverable.

The TM FORUM invites any party to contact the TM FORUM Team Administrator if it is aware of a claim of ownership of any patent claims that would necessarily be infringed by implementations of this TM FORUM Standards Final Deliverable by a patent holder that is not willing to provide a license to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this TM FORUM Standards Final Deliverable. TM FORUM may include such claims on its website but disclaims any obligation to do so.

TM FORUM takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this TM FORUM Standards Final Deliverable or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any effort to identify any such rights. Information on TM FORUM's procedures with respect to rights in any document or deliverable produced by a TM FORUM Collaboration Project Team can be found on the TM FORUM website. Copies of claims of rights made available for publication and any assurances of licenses to be made available, or the result of an attempt made to obtain a general license or permission for the use of such proprietary rights by implementers or users of this TM FORUM Standards Final Deliverable, can be obtained from the TM FORUM Team Administrator. TM FORUM makes no representation that any information or list of intellectual property rights will at any time be complete, or that any claims in such list are, in fact, Essential Claims.

Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

Table of Contents

Notice	2
Table of Contents	4
1. Overview	5
2. eTOM Processes and SID Data Entities	6
2.1. eTOM business activities	6
2.2. SID ABEs	8
2.3. eTOM L2 - SID ABEs links	9
2.4. Functional Framework Functions	9
3. TMF OPEN APIs & Events	11
3.1. Exposed APIs	11
3.2. Dependant APIs	12
3.3. Events	15
4. Machine Readable Component Specification	16
5. References	17
5.1. TMF Standards related versions	17
5.2. Jira references	17
5.3. Further resources	17
6. Administrative Appendix	18
6.1. Document History	18
6.1.1. Version History	18
6.1.2. Release History	19
6.1.3. Acknowledgments	19

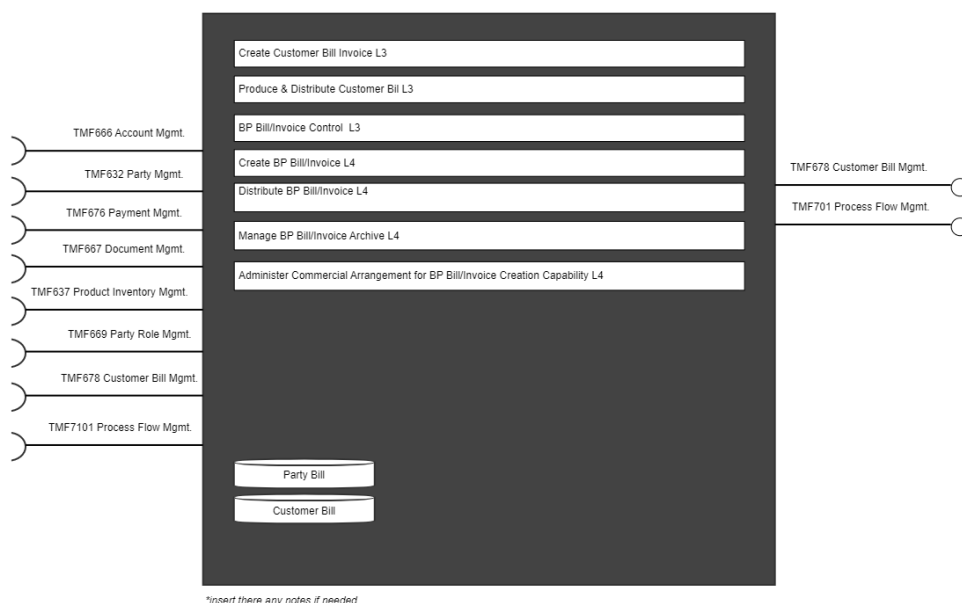
1. Overview

Component Name	ID	Description	ODA Function Block
Bill Generation Management	TMFC030	Bill generation management .manages the party invoice management. It addresses the invoice formatting, presentation and dispatching to the proper means of communication.	Party Management

2. eTOM Processes and SID Data Entities

2.1. eTOM business activities

TMFC030 : Bill Generation Management



eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.3.9	L2	Customer Bill Invoice Management	Ensure the bill invoice is created, physically and/or electronically produced and distributed to customers, and that the appropriate taxes, discounts, adjustments, rebates and credits for the products and services delivered to customers have been applied.
1.3.9.2	L3	Create Customer Bill Invoice	Production of a timely and accurate invoice in accordance with the specific billing cycles and reflective of the final charges for services, together with any adjustments, delivered to the customer by the Service Provider and respective other parties.
1.3.9.2.1	L4	Render & Format Customer Invoice	Render and format the customer bill invoice.
1.3.9.2.2	L4	Deliver Electronic Invoice	Deliver the electronic copy of an invoice to customers.

Identifier	Level	Business Activity Name	Description
1.3.9.2.3	L4	Verify Customer Invoice Quality	Verify Customer invoice quality before distribution to the customer in electronic form and the process responsible for physical invoice production and distribution.
1.3.9.2.4	L4	Manage Customer Invoice Quality Archive	Store the customer invoice for a period of time is to address regulation and/or internal requirements, during which they can be accessed to support any customer or regulator agency inquiries on bill invoices.
1.3.9.3	L3	Produce & Distribute Customer Bill	Physical production and distribution of bills to customers in accordance with the specified billing cycle.
1.3.9.3.1	L4	Co-ordinate Billing Insertion	Co-ordinate with promotional processes for any billing insertions to be included with the bill.
1.3.9.3.2	L4	Establish & Manage Bill Production Cycle	Establish and manage the physical bill production cycle.
1.3.9.3.3	L4	Deliver Invoice Information	Deliver the invoice information to the physical production processes.
1.3.9.4	L3	Pricing, Discounting, Adjustments & Rebates Application	Ensure that the bill invoice is reflective of all the commercially agreed billable events and any bill invoice adjustments agreed between a Service Provider and the customer.
1.3.9.4.2	L4	Apply Pricing, Discounting, Adjustments & Rebates to Customer Account	Determine the customer account or customer specific pricing, charges, discounts, and taxation that should be delivered to the invoice(s) for the customer.
1.6.15	L2	BP Bill/Invoice Management	Business Partner Bill/Invoice Management manages the Business Partner bill/invoice process, controls bills/invoices, manages the lifecycle of bills/invoices. A bill is a notice for payment which is supposed to be preceded by an invoice in most cases.
1.6.15.2	L3	BP Bill/Invoice Control	Establish and maintain Business Partner bill invoice formats, maintain lists of parties who are eligible for receiving bills/invoices, and define the billing cycles.
1.6.15.2.1	L4	Establish & Maintain BP Bill Invoice Format	Establish and maintain BP bill invoice formats, and any interaction with specific parties to modify the format.
1.6.15.2.2	L4	Maintain Bill Invoice BP List	Maintain lists of parties who are eligible for receiving bills/invoices.

Identifier	Level	Business Activity Name	Description
1.6.15.2.3	L4	Define BP Billing Cycle	Define the billing cycles and their dates according to cash flow needs as established by financial management processes.
1.6.15.3.2	L4	Create BP Bill/Invoice	Produce a timely and accurate bill/invoice in accordance with a specific billing cycle, on demand after the purchase of an offering, on request by a BP, and so forth. Ensure that a bill/invoice is reflective of the final charges for products, together wit
1.6.15.3.2	L4	Create BP Bill/Invoice	Produce a timely and accurate bill/invoice in accordance with a specific billing cycle, on demand after the purchase of an offering, on request by a BP, and so forth. Ensure that a bill/invoice is reflective of the final charges for products, together wit
1.6.15.3.3	L4	Distribute BP Bill/Invoice	Provide bills/invoices to one or more parties and ensure the delivery of bills/invoices to one or more parties.
1.6.15.3.4	L4	Manage BP Bill/Invoice Archive	Store a BP bill/invoice for a period of time to address regulation and/or internal requirements, during which it can be accessed to support any BP, such as a government/regulator agency or internal BP, inquiries about a bill/invoice.
1.6.15.3.5	L4	Receive BP Bill/Invoice	Receive and record the bill/invoice from a BP. Compare a BP bill/invoice against all transactions with the BP that would result in a bill/invoice being sent to the enterprise. Manage the interactions between a BP and an enterprise. Approve a BP bill/invo
1.6.15.3.6		Administer Commercial Arrangement for BP Bill/Invoice Creation Capability	Establish the requirements for, and manage the agreed commercial arrangements with, appropriate outsourced parties of the creation capabilities.

2.2. SID ABEs

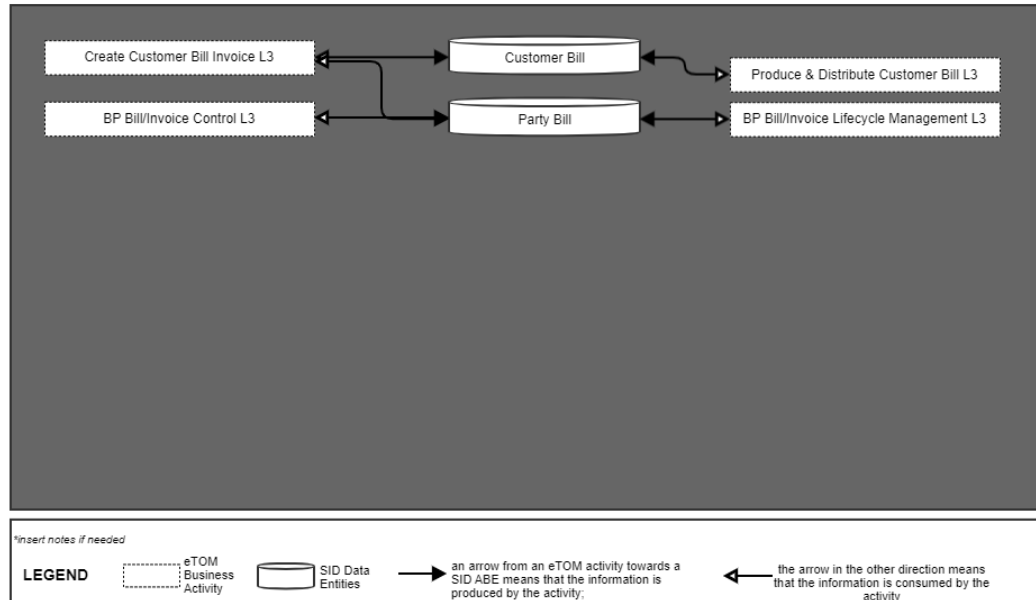
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Party Bill ABE	n/a
Customer Bill ABE	n/a

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Sub-Domain Functions Level 1	Sub-Domain Functions Level 2
62	Invoice Items Listing	Invoice Items Listing lists all invoice items for a specific invoice.	Invoice Management	Invoicing
63	Invoice Listing	Invoice Listing function will list all invoices for a customer both over time and for customers with multiple invoices.	Invoice Management	Invoicing
65	Bill Image Presentation	Bill Image Presentation provides presentation of an exact bill image or after invoking a transactional document generation function.	Invoice Management	Invoicing
309	Invoice Balance Calculation	Provides the means to calculate the balance due for an invoice/bill.	Invoice Management	Invoicing

Function ID	Function Name	Function Description	Sub-Domain Functions Level 1	Sub-Domain Functions Level 2
310	Invoice Charges Compilation	Invoice Charges Compilation assembles charges (including charge distribution- charges incurred by other customers), credits, taxes, fees and adjustments that affect the balance due.	Invoice Management	Invoicing
312	Invoice Detail Collection	Provides appropriate levels of detail regarding items on the invoice. This detail is provided to revenue reporting and/or Bill Format &; Render.	Invoice Management	Invoicing
311	Invoice Totals Calculation	Provides subtotals and totals at various levels.	Invoice Management	Invoicing
329	Invoice Tax Calculation	"Invoice Tax Calculation provides the necessary functionality to calculate taxes, including surcharges and fees, where applicable. This function can occur within the Invoicing application or through the use of an external Tax module."	Invoice Management	Invoicing

3. TMF OPEN APIs & Events

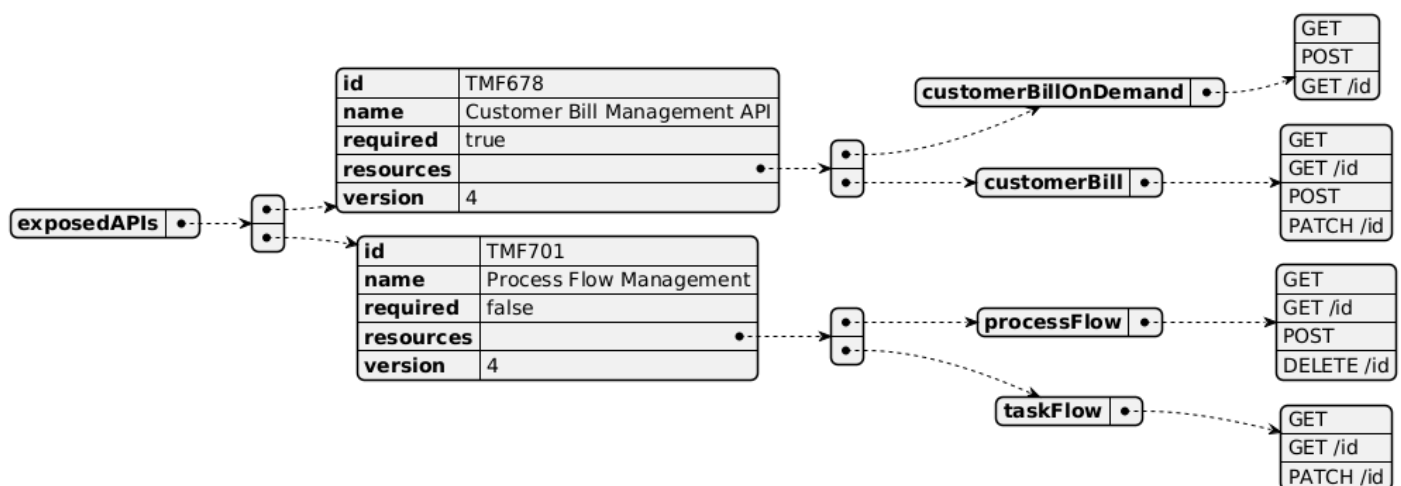
The following part covers the APIs and Events; This part is split in 4:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

<Note to be inserted into ODA Component specifications: If a new Open API is required, but it does not yet exist. Then you should include a textual description of the new Open API, and it should be clearly noted that this Open API does not yet exist. In addition a Jira epic should be raised to request the new Open API is added, and the Open API team should be consulted. Finally, a decision is required on the feasibility of the component without this Open API. If the Open API is critical then the component specification should not be published until the Open API issue has been resolved. Alternatively if the Open API is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

3.1. Exposed APIs

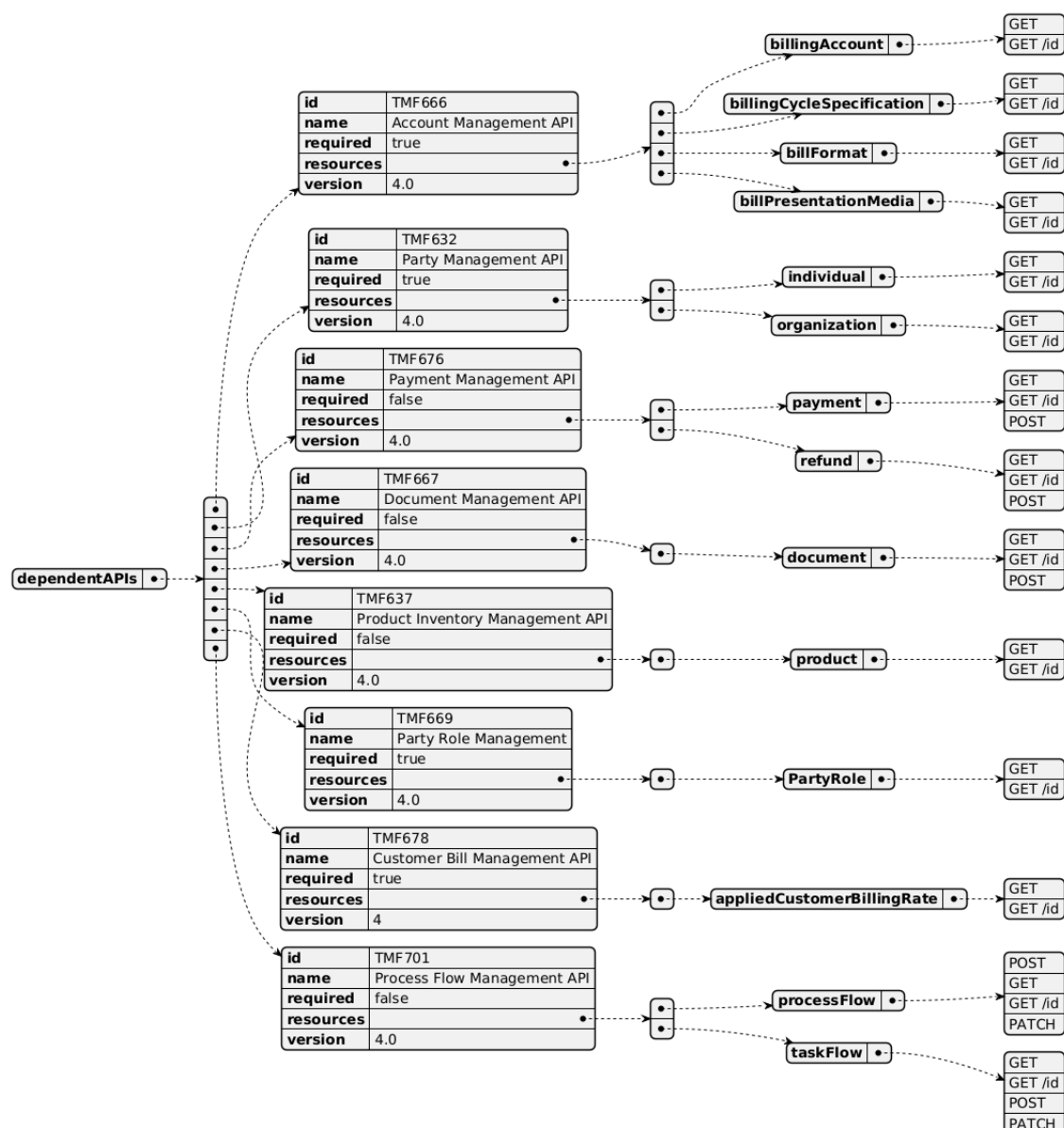
Following diagram illustrates API/Resource/Operation:



API ID	API Name	Mandatory / Optional	Operations
TMF678	Customer Bill Management	Mandatory	customerBillOnDemand • GET • GET /id • POST customerBill • GET • GET /id • POST • PATCH /id :
TMF701	Process Flow Management	Optional	processFlow: • GET • GET /id • POST • DELETE /id taskFlow: • GET • GET /id • PATCH /id

3.2. Dependant APIs

Following diagram illustrates API/Resource/Operation potentially used by the product catalog component:



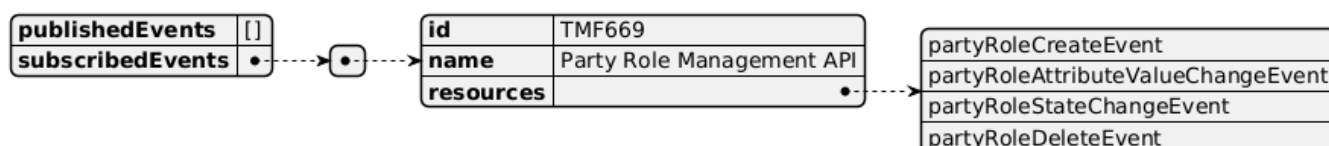
API ID	API Name	Mandatory / Optional	Operations
TMF666	Account Management API	Mandatory	billingAccount: - GET - GET /id billingCycleSpecification: - GET - GET /id billFormat:

API ID	API Name	Mandatory / Optional	Operations
			• GET • GET /id billPresentationMedia: • GET • GET /id
TMF632	Party Management API	Optional	individual: • GET • GET /id organization: • GET • GET /id
TMF676	Payment Management API	Optional	payment: • GET • GET /id • POST refund: • GET • GET /id • POST
TMF667	Document Management API	Optional	document: • GET • GET /id • POST
TMF637	Product Inventory Management API	Optional	product: • GET • GET /id
TMF669	Party Role Management API	Optional	partyRole: • GET • GET /id

API ID	API Name	Mandatory / Optional	Operations
TMF678	Customer Bill Management API	Mandatory	appliedCustomerBillingRate: - GET - GET /id
TMF701	Process Flow Management API	Optional	processFlow: - POST - GET - GET /id - PATCH /id taskFlow: - GET - GET /id - POST - PATCH /id

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



The type of event could be:

- Create : a new resource has been created (following a POST).
- Delete: an existing resource has been deleted.
- AttributeValueChange: an attribute from the resource has changed - event structure allows to pinpoint the attribute.
- InformationRequired: an attribute should be valued for the resource preventing to follow nominal lifecycle - event structure allows to pinpoint the attribute.
- StateChange: resource state has changed.

4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

5.2. Jira references

- **SID:** to be updated to separate Customer Bill ABE from Party Bill ABE  [ISA-847](#) - TMFC030: Bill Generation Management - SID update request **BACKLOG**
- **eTOM:** to review mapping L3/L4 mapping for: FX-1225
 - Produce & Distribute Customer Bill
 - Pricing, Discounting, Adjustments & Rebates Application
 - BP Bill/Invoice Management
- **OpenAPI:** TMF678_Customer_Bill_Management remove following resources:
 [AP-4648](#) - TMFC030: Bill Generation Management **DONE**
 - Bill Cycle
 - Bill Cycle Specification
- **ODA Component:** add new component for Party Communication Mgmt.

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	27 Sep 2023	Gaetano Biancardi	Final edits prior to publication
2.0.0	21 Nov 2023	Gaetano Biancardi	Open API: Functional Framework: • 309 Invoice Balance Calculation > added • 310 Invoice Charges Compilation > added • 312 Invoice Detail Collection > added • 311 InvoiceTotals Calculation > added Open API: • ○ TMF666 > removed "appliedCustomerBillingRate" resource and related API methods
2.1.0	19 May 2024	Gaetano Biancardi	updated to new component template updated to odf v.23.5 removed TMF672 as mandatory dependent API Exposed API, removed: TMF688 Event Mgmt TMF672 User Roles and Permissions Dependant API, removed TMF688 Event Mgmt
2.1.0	12 Jul 2024	Amaia White	Final edits prior to publication
2.2.0	21 Oct 2024	Gaetano Biancardi	Dependant API:

Version Number	Date Modified	Modified by:	Description of changes
			- Added TMF678 Customer Bill Mgmt. to retrieve appliedCustomerBillingRate Exposed API: - added POST to TMF678 Customer Bill resource
2.2.0	03 Dec 2024	Elisabeth Andersson	Updated YAML file to remove TMF688 and TMF672 from the core specification, as the two APIs are moved to supporting functions
2.2.0	27 Dec 2024	Rosie Wilson	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	20 Oct 2023	Amaia White	Initial release of document
Pre-production	20-Nov-2023	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	20-Dec-2023	Amaia White	New version release 2.0
Production	09-Feb-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New version release 2.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	27 Dec 2024	Rosie Wilson	Updated to version 2.2.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.1.3. Acknowledgments

Team Member (@mention)	Company	Role*
Gaetano Biancardi	Accenture	Editor
Sylvie Demarest	Orange	Reviewer
Elisabeth Andersson	Matrixx	Additional Input

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer